

PART 1: DOWNLOAD THE CHOSEN SPECIES FROM NCBI AND UPLOAD IT IN GALAXY FOR GENOME EXPLORATION

SPECIES: *Hipposideros armiger* (Great Roundleaf Bat) genome assembly (GCF_001890085.2 / ASM189008v1)

GALAXY FILE NAME: 1: *Hipposideros_armiger_genome_original.fna.gz*

NOTE: This is only a screenshot as I cannot upload the raw file on GitHub due to its large file size. Therefore, this screenshot will serve as evidence that I indeed downloaded the raw genome.

The screenshot displays the Galaxy web interface. At the top, a yellow folder icon represents the uploaded dataset. Below it, the dataset name is shown: **GCF_001890085.2_ASM189008v2_genomic.fna**. A 'Share' button is visible. The 'Details' section shows the file type as 'Compressed Archive Folder', size as '521 MB', and the location as 'CitIsen/winah marie\Downloads' with a timestamp of '12/06/2026 8:37 am'. A 'Properties' button is at the bottom of the details section.

The main workspace shows the dataset '1: *Hipposideros_armiger_genome_original.fna.gz*' with a status of 'ok'. It indicates '7,387 sequences' in 'fasta.gz' format, with a size of '637.6 MB'. A warning message states: 'This dataset is large and only the first 100 KB is shown below.' Below this, a preview of the FASTA sequence is visible, starting with:
>NW_017731350.1 *Hipposideros armiger* isolate ML-2016 unplaced genomic scaffold, ASM189008v1 scaffold_0, whole genome shotgun sequence
AACGACAAGCTTGCTGAATTATGGATTATGTAACGTATTAAGTAATTGcccttctccctc
GCGAATTTTGTGTTCTTGCTTCCCTCTGATCCAGTCCCCTCCCTAATACTTTACAA/
TATTTCTGTCAGATCAGAAGAGTCTCAGTGCAGAGATTAGGGTGATCGGGAATGCTCA
GCAATTTCTCAGAGATCAGAGGGATGCAAGTTTCTAGAGGAAAGcaaccattatattatag
AGAGCTAAAGTCTCAGTCCCTGGACAGGACCGTTGTTGATTGCGCTTCTCTTAA/
GAGACCGAGTGCCTGGCAGATAATGTGATCAGATGAGGTTGCGCTTTCAGACCTGGI
CCTTCCCGGCGCCACCAAGCTTCGATTTGTGGTTTGAAGTGAAGTTCAACAGCAGGI
TCTTTAGTTGaatctcttctctccattCAAAGAGAAAATTTACCAGCAATTTACTAGCTTTT
TCAAGCAATACCTGGCACGGAGTATCAGACTTGACGTTAGATAAGGTAATGATGGGGA
ACATGGTGCTAACATTATCGCACATGattatgccaggtactcACGgataattcttatttaccittacaAC
TGCAAAAGATGAGTATTTAATCTCTGTTTTGCAAATGATGCTGTGCGGGGTTACGTGA
GTAAGTGGCTGAAGTGGATTGTTTCCCTTACAGGGCTTAGGTTTGAACACCTGAA
TGTGCTTCTTCCACGCTATTataagggaaaaaatatttttaaaaaagacacttggggccgagtggtcagtt
gggttaagagctcattcaagggtgcagctctgagcaacaggtcccggttcggtccgccaalccgagctgccaac

The left sidebar shows the 'Activities' panel with 'All Tools' selected. The 'Tools' section includes 'Discover 2k+ Tools', 'search tools', 'Get Data', 'Send Data', 'Collection Operations', 'General Text Tools', 'Text Manipulation', 'Filter and Sort', 'Join, Subtract and Group', 'Datamash', 'Genomic File Manipulation', 'FASTA/FASTQ', 'FASTQ Quality Control', 'SAM/BAM', 'BED', 'VCF/BCF', and 'Nanopore'. The right sidebar shows the 'History' panel with a search bar and a list of datasets, including 'Genome_Exploration_II_Suan_Hipposideros_ar...', 'Default Storage', '12 GB', and a list of datasets: '6: Hippositeros_armiger_filter ed_10kb', '5: Hipposideros_armiger_Computer sequence length (sorted)', '3: Hippositeros_armiger_Compute sequence length', '2: Hippositeros_armiger_Assembly_Statistics_gfastats', and '1: Hipposideros_armiger_genome_original.fna.gz'.

Figure 1. NCBI Assembly page showing accession GCF_001890085.2 and Galaxy History confirming successful upload of the *Hipposideros armiger* genome assembly file.